

SMART INDIA HACKATHON 2026

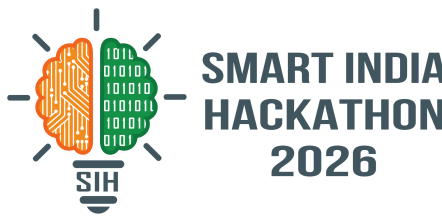
- **Problem Statement ID – 26171**
- **Problem Statement Title - On-device Visual Perception for Light-weight Browser Agents**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 150126**
- **Team Name- Redacted**





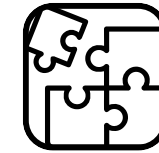
EDWARD

Enterprise **D**ata **W**atchdog for **A**nonymity, **R**isk & **D**efense



What is it?

A **browser extension** that sits between the webpage and any AI agent backend. It reads the page on your device, **replaces PII** with tokens ([PII_NAME_1]) and **redacts faces, QR codes and text in images**. Only this sanitized view is sent out. The actions that come back run locally, and the tokens are swapped back to the real values there.



Proposed Solution / Approach

- All perception **runs on the device** (DOM plus images), so raw PII never crosses the network.
- A **lightweight agent** gets a compact, structured view of the page instead of screenshots.



Innovation and uniqueness

- **Fail-closed:** anything uncertain or unreadable is redacted, never passed through.
- **Secret fields are never read:** passwords, OTPs and card details.
- **Token egress policy:** real values can't be leaked through the agent's actions.
- **Works with any backend:** Groq, any OpenAI-compatible server or a custom server, with no code changes.
- **Model tiers matched to the hardware:** WebGPU or WASM, chosen automatically.

Images, full size: original vs sent

Call Priya: +91 98765 43210
priya.sharma.canary@example.com

Original (stays on the laptop)

Call [REDACTED]
[REDACTED]

Sent to the backend: redacted, downscaled JPEG

Order summary
Status: shipped to the front desk
PAN: ABCPE1234F
Phone: +91 98765 43210

Original (stays on the laptop)

Order summary
Status: shipped to the front desk
PAN [REDACTED]
Phone: [REDACTED]

Sent to the backend: redacted, downscaled JPEG

EDWARD

TECH STACK

EDWARD

TS

TypeScript

Strict application language

W

WXT

Browser extension framework

Vite

Build tooling

</>

Manifest V3

Extension platform standard

Chrome

Chromium browser target

Firefox

Gecko browser target

MediaPipe Tasks Vision

BlazeFace face detection

ONNX Runtime Web

SCRFD face detection

Transformers.js

BERT-small NER / PII detection

T

Tesseract.js

OCR for text in images

ZXing-WASM

QR and barcode detection

Regex rules

Deterministic PII detection

#

Verhoeff

ID-number checksum validation

Luhn

Card-number checksum validation

Token Map

In-memory redaction mapping

Image Redaction

Local image sanitization

Network Block

Blocks model-code network access

OpenRouter

Multi-LLM provider

Qwen

Qwen3.5-27B Multimodal reasoning model

OpenAI-compatible API

Standard LLM API interface

Custom Agent Server

Alternative backend

Extension Storage

Settings storage

IndexedDB

Metrics logs and image cache

Vitest

Unit testing

Playwright

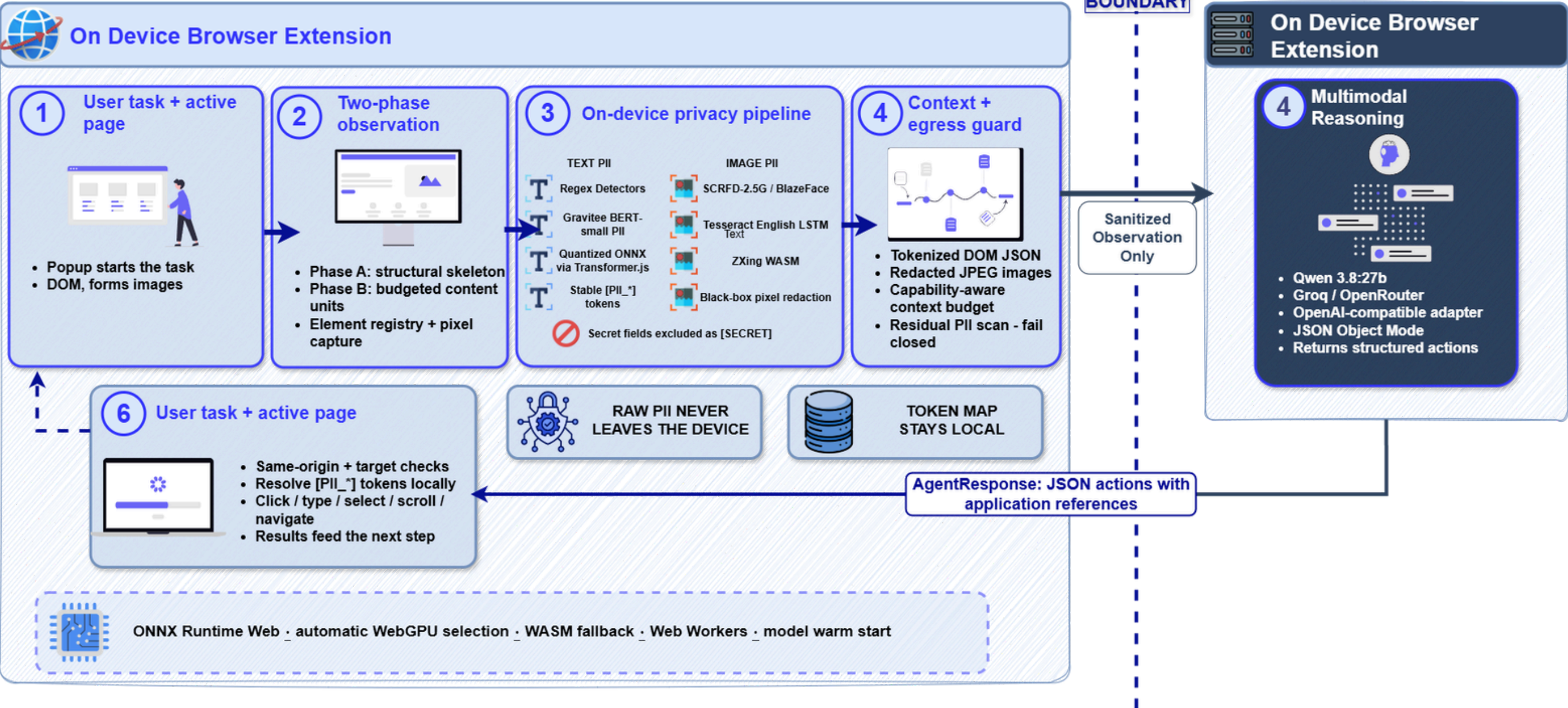
End-to-end testing

ESLint

Static analysis and module boundaries

web-ext

Firefox testing and linting



FEASIBILITY

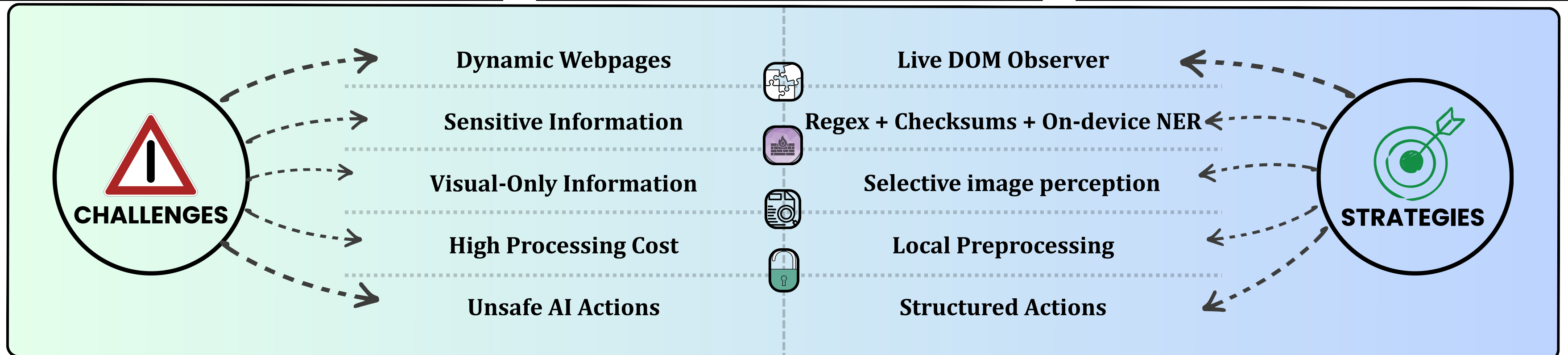
- Uses browser-native APIs for DOM, accessibility and page geometry
- Performs PII detection locally using regex, checksums and on-device NER
- Uses a modular architecture with independent compute, model and backend layers
- Uses IndexedDB without storing raw PII or raw image bytes
- Prototype validated with 310/310 unit tests and 11/11 E2E tests

VIABILITY

- Reduces data exposure by sanitizing PII before network transmission
- Preserves functionality by replacing sensitive values with stable tokens
- Supports backend-agnostic reasoning, allowing different AI backends
- Blocks unsafe egress instead of allowing sensitive tokens to leave the browser
- Enables practical browser automation while keeping sensitive data local

WHY IT WORKS

- Privacy is enforced before AI reasoning, not after
- Uses defense-in-depth: detection + tokenization + egress controls
- Uses fail-closed security for unsafe actions and data leakage
- Uses structure-first processing to minimize unnecessary computation
- Keeps the architecture extensible and model-agnostic





Economic Benefits

Lets enterprises adopt browser-automation AI agents for form-filling and data entry without the legal or compliance risk of sending customer PII to a third-party LLM API — avoiding costly data-leak liability while still gaining automation productivity.



Social Benefits

Protects citizens' personal data — Aadhaar, PAN, phone numbers, addresses, faces — whenever they use an AI assistant on government or institutional portals, building public trust in AI-assisted digital services.

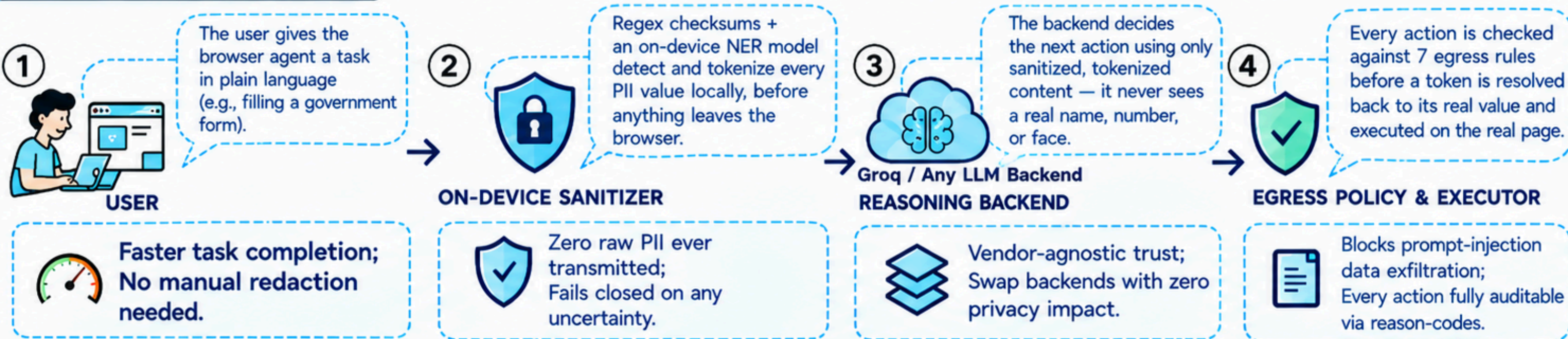


Institutional & Governance Benefits

Backend-agnostic and auditable by design — no vendor lock-in to any single AI provider, full reason-code logging for compliance review, and a defense-in-depth architecture regulators can inspect.

STAKEHOLDERS & IMPACTS

Sample Scenario



Institutional & Governance Impact

- Architecture supports data-protection compliance for citizen-facing AI deployments.
- Backend-agnostic design avoids vendor lock-in for public-sector AI adoption.

OUR PROMISE

Our solution perfectly aligns with the UN Sustainable Development Goals.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



16 PEACE, JUSTICE AND STRONG INSTITUTIONS



Verified, Not Estimated:

- ✓ **310/310** unit tests passing.
- ✓ **11/11** end-to-end tests passing, stable across 6 consecutive full-suite runs.
- ✓ **100%** PII canary recall — zero leaks in real on-device inference.

$$\text{Recall} = \frac{\text{Detected PII}}{\text{Total PII}} = 100\%$$

***Consolidated proof documents of all testing are provided in the appendix.**



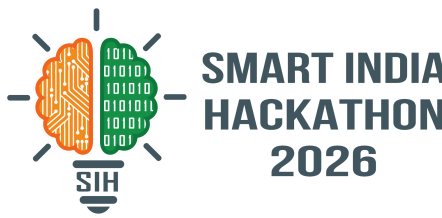
Edward's egress policy is validated against a real prompt-injection attack fixture — the extension provably never navigates with a leaked token.



0% Zero PII leaks recorded across all automated test runs — this is why trusting the reasoning backend is unnecessary.



RESEARCH AND REFERENCES



On-Device Vision & Perception Models

- **BlazeFace (Bazarevsky et al., 2019)**: Sub-millisecond neural face detection architecture for mobile GPUs. <https://arxiv.org/abs/1907.05047>
- **SCRFD (Guo et al., ICLR 2022)**: Sample and Computation Redistribution for Efficient Face Detection (Tier-2 face detection). <https://arxiv.org/abs/2105.04714>

PII Detection Model

- **Bert-small-pii-detection (gravitee-io, Hugging Face)**: Fine-tuned BERT-small NER model, 25 PII entity labels, ONNX/quantized for browser inference. <https://huggingface.co/gravitee-io/bert-small-pii-detection>
- **Set-of-Mark Prompting (Yang et al., 2023)**: Visual grounding technique that unlocks strong grounding ability in GPT-4V. <https://arxiv.org/abs/2310.11441>

OCR & Barcode

- **Tesseract OCR (LSTM engine, v4+)**: Open-source line-recognition OCR engine used via Tesseract.js. <https://github.com/tesseract-ocr/tesseract>
- **ZXing ("Zebra Crossing") Project**: Open-source barcode/QR image processing library, used via zxing-wasm. <https://github.com/zxing/zxing>

Legal & Regulatory Sources

- **The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)**: Official Government of India legislation on digital personal data processing, notified by the Ministry of Electronics and Information Technology, August 2023. https://www.trai.gov.in/sites/default/files/2024-09/DPDT_11082024.pdf

Runtimes / Inference

- **ONNX Runtime Web**: In-browser WASM/WebGPU execution engine for SCRFD and the NER model. <https://onnxruntime.ai/docs/tutorials/web/>
- **Transformers.js (Hugging Face)**: Runs the token-classification (NER) pipeline client-side, no server call. <https://huggingface.co/docs/transformers.js>
- **MediaPipe Tasks Vision (Google)**: Face-detector task API wrapping BlazeFace for in-browser inference. https://ai.google.dev/edge/mediapipe/solutions/vision/face_detector

Checksum Algorithms

- **Verhoeff Algorithm**: Checksum scheme used to validate Aadhaar-number matches before redaction. https://en.wikipedia.org/wiki/Verhoeff_algorithm
- **Luhn Algorithm**: Checksum scheme used to validate card-number matches before redaction. https://en.wikipedia.org/wiki/Luhn_algorithm

Standards & Law

- **Manifest V3 (Chrome / Firefox)**: Extension platform standard Edward is built against. <https://developer.chrome.com/docs/extensions/develop/migrate/what-is-mv3>
- **WebGPU Specification (W3C)**: Browser compute standard Edward uses when available, falling back to WASM. <https://www.w3.org/TR/webgpu/>
- **The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)**: Official Government of India legislation on digital personal data processing, notified by MeitY, August 2023. <https://www.meity.gov.in/data-protection-framework>

Our Project Repository

- **GitHub repo (SIH26171 / Edward)**: <https://github.com/rahman-misbah/SIH26171>